THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 6: **We Distribute, Yet Things Multiply** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $x + y = 9$ and $xy = 14$, then $x^2 + y^2$ equals: 1

(a) 67 **(b)** 81 **(c)** 53 **(d)** 109

2. If $x - y = 3$ and $xy = 20$, then $x^2 + y^2$ equals: 1

(a) 49 **(b)** -31 **(c)** 9 **(d)** 29

3. The value of $512^2 - 488^2$ is: 1

(a) 576 **(b)** 500288 **(c)** 24100 **(d)** 24000

4. If $x + \frac{1}{x} = 3$, then $x^2 + \frac{1}{x^2}$ equals: 1

(a) 8 **(b)** 7 **(c)** 11 **(d)** 9

5. If $(x + p)(x + q) = x^2 + 7x + 12$, then $p^2 + q^2$ equals: 1

(a) 73 **(b)** 37 **(c)** 25 **(d)** 49

6. The area of a square of side 25 cm exceeds that of a square of side 22 cm by: 1

(a) 484 **(b)** 9 **(c)** 75 **(d)** 141

7. The value of $24^2 - 23^2$ is: 1

(a) 24 **(b)** 47 **(c)** 49 **(d)** 48

8. $(a + b)^2 = a^2 + b^2$ is true only when: 1

(a) $a = b$ **(b)** $ab = 0$ **(c)** $a = 1$ **(d)** $a + b = 0$

9. If $x + y = 11$ and $xy = 28$, then $x^2 + y^2$ equals: 1

(a) 65 **(b)** 93 **(c)** 177 **(d)** 121

10. If $x - y = 4$ and $xy = 45$, then $x^2 + y^2$ equals: 1

(a) -74 **(b)** 16 **(c)** 106 **(d)** 61

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. If $x + y = 12$ and $xy = 35$, find $x^2 + y^2$. 2

12. Evaluate $612^2 - 388^2$ using an identity. 2

13. If $(x + p)(x + q) = x^2 + 8x + 15$, find $p^2 + q^2$. 2

14. If $x + y = 8$ and $xy = 15$, find $x^2 + y^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If $x - y = 4$ and $xy = 21$, find $x^2 + y^2$ and $(x + y)^2$. 3

16. Simplify $(3x + 2)^2 - (3x - 2)^2$. 3

17. If $a + b = 10$ and $ab = 24$, find $a^2 + b^2$ and $(a - b)^2$. 3

OR

If $x - y = 6$ and $xy = 40$, find $x^2 + y^2$ and $(x + y)^2$. 3

18. Simplify $(4x + 5)^2 - (4x - 5)^2$. 3

OR

If $a + b = 13$ and $ab = 40$, find $a^2 + b^2$ and $(a - b)^2$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The sides of a rectangle are $(3x + 2)$ m and $(3x - 2)$ m. Find its area as an expression, its value at $x = 5$, and by how much it is less than the area of a square of side $3x$ m. 5

OR

If $x + y = 13$ and $x^2 + y^2 = 97$, find xy, $(x - y)^2$ and $x - y$ (take $x > y$). 5

20. The sides of a rectangle are $(4x + 3)$ m and $(4x - 3)$ m. Find its area as an expression, its value at $x = 6$, and by how much it is less than the area of a square of side $4x$ m. 5

OR

If $x + y = 16$ and $x^2 + y^2 = 146$, find xy, $(x - y)^2$ and $x - y$ (take $x > y$). 5

END OF QUESTION PAPER — Total: 40 Marks

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